

Structural Dynamics : Course project

Reduced Cooling Fan Vibration Through Mass Balancing



by

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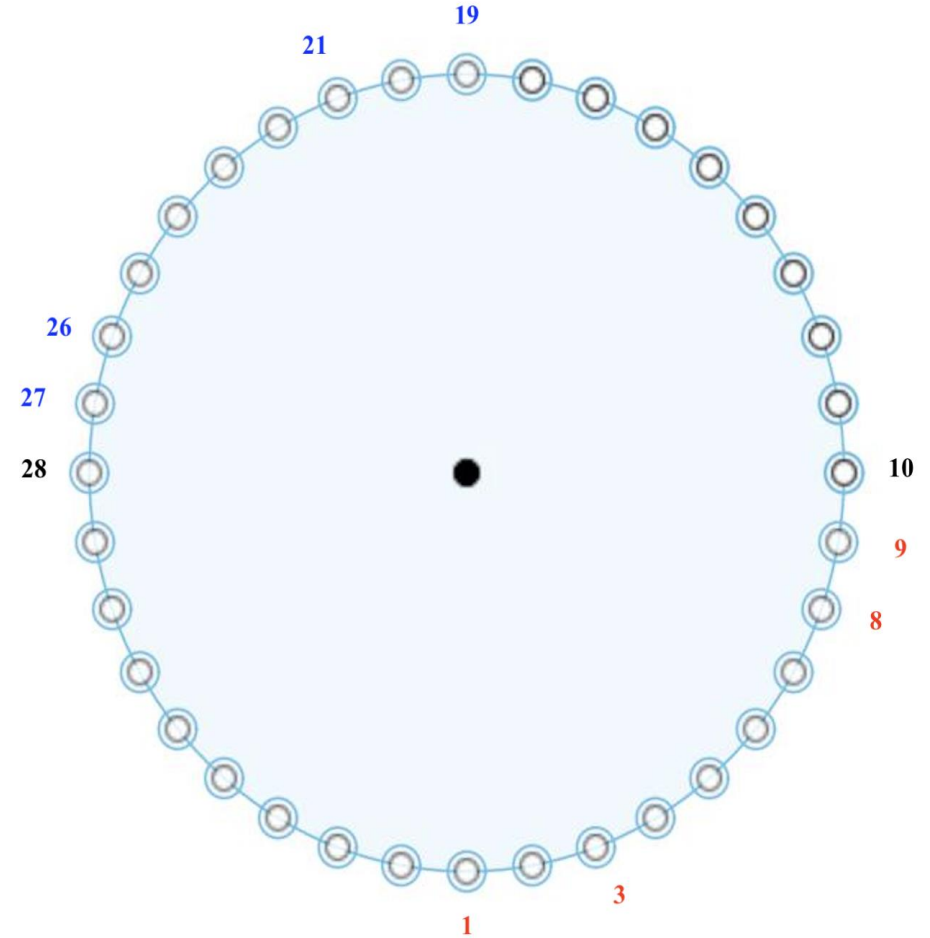
Introduction

- In this cooling fan, the motor rotates the fan blades. However, due to four broken blades, an unbalanced force is created, causing the entire cooler body to move in a circular motion, specifically in a circular path.



Unbalanced Forced Analysis

- No. of Blade =36
- Angle b/w adjacent blades =10 degree

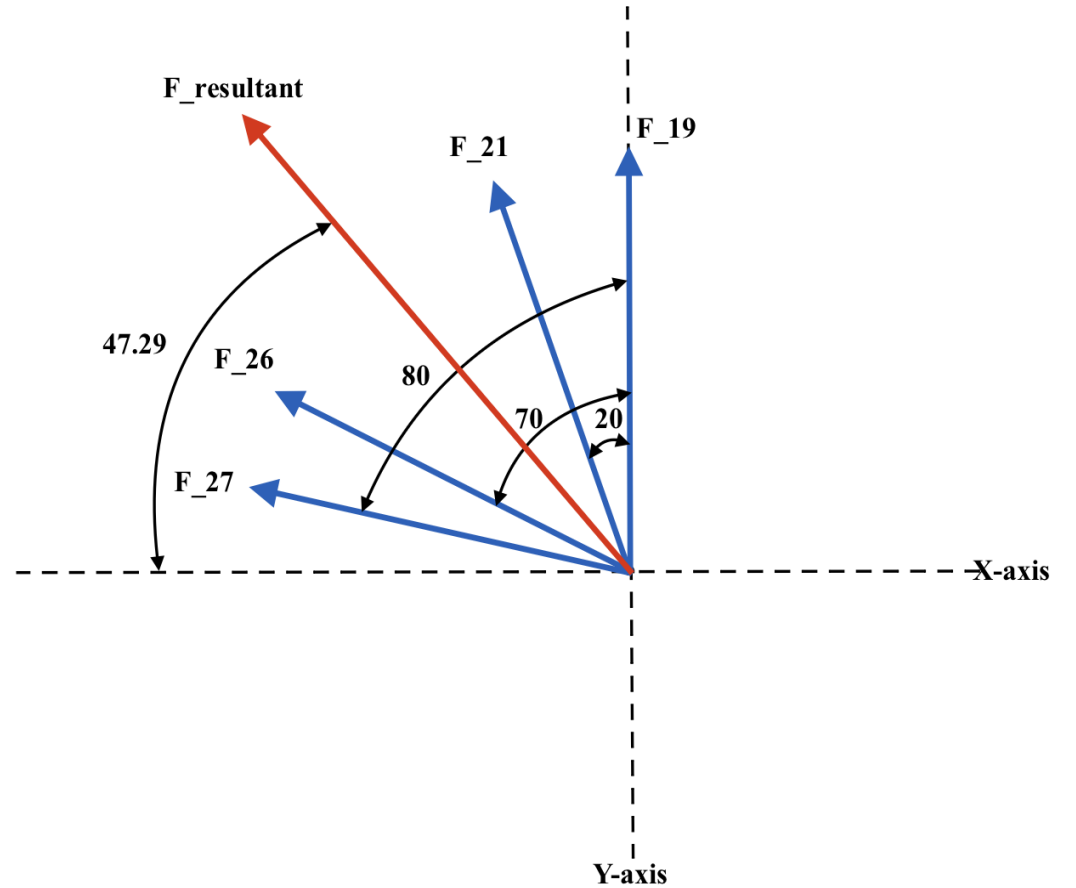


Blades broken - 1, 3, 8, 9

Blades to be removed - 19, 21, 26, 17

Force Vector Diagram

- $F_x = -2.266 \cdot F \text{ (N)}$
- $F_y = 2.455 \cdot F \text{ (N)}$
- $F_{\text{resultant}} = 3.340 \cdot F \text{ (N)} = 5.419 \text{ N}$
- $\Theta_{\text{resultant}} = 47.29^\circ$
- $m = 1.51 \text{ gram}$
- $e = 5 \text{ cm}$
- $w = 146.6 \text{ rad/sec}$
- $F = 1.6226 \text{ N}$
- Weight of cooler = 24 N



Data Acquisition(unbalanced fan)

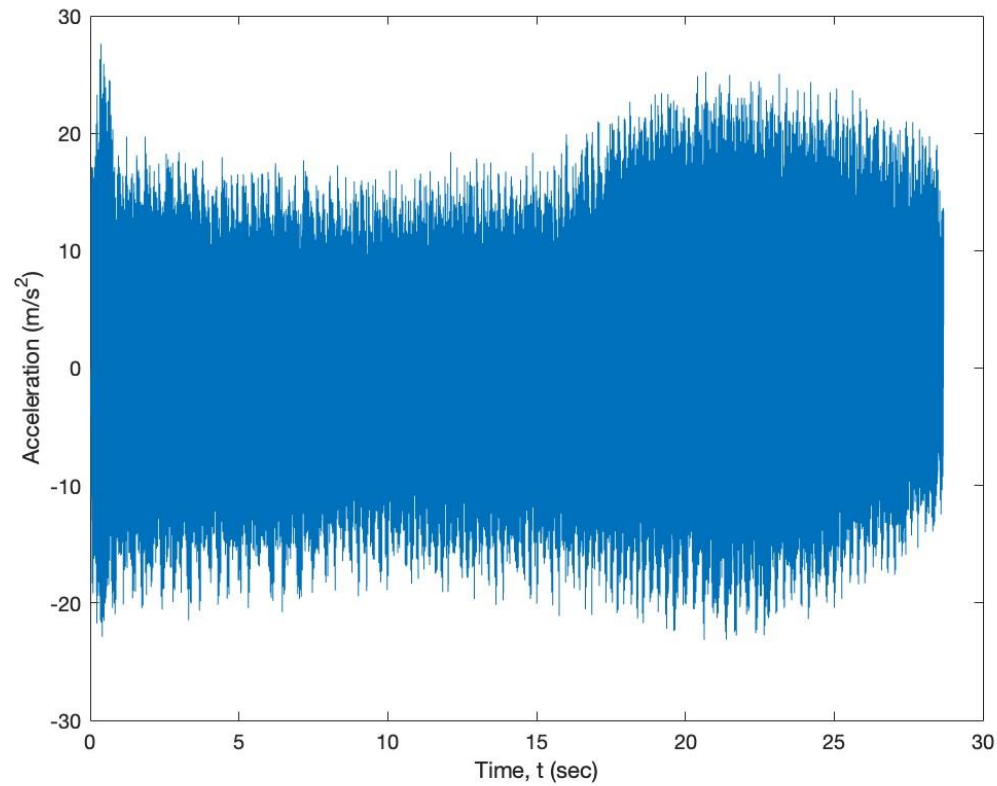


Fig : plot of Acceleration in x (m/s^2) Vs Time(sec)

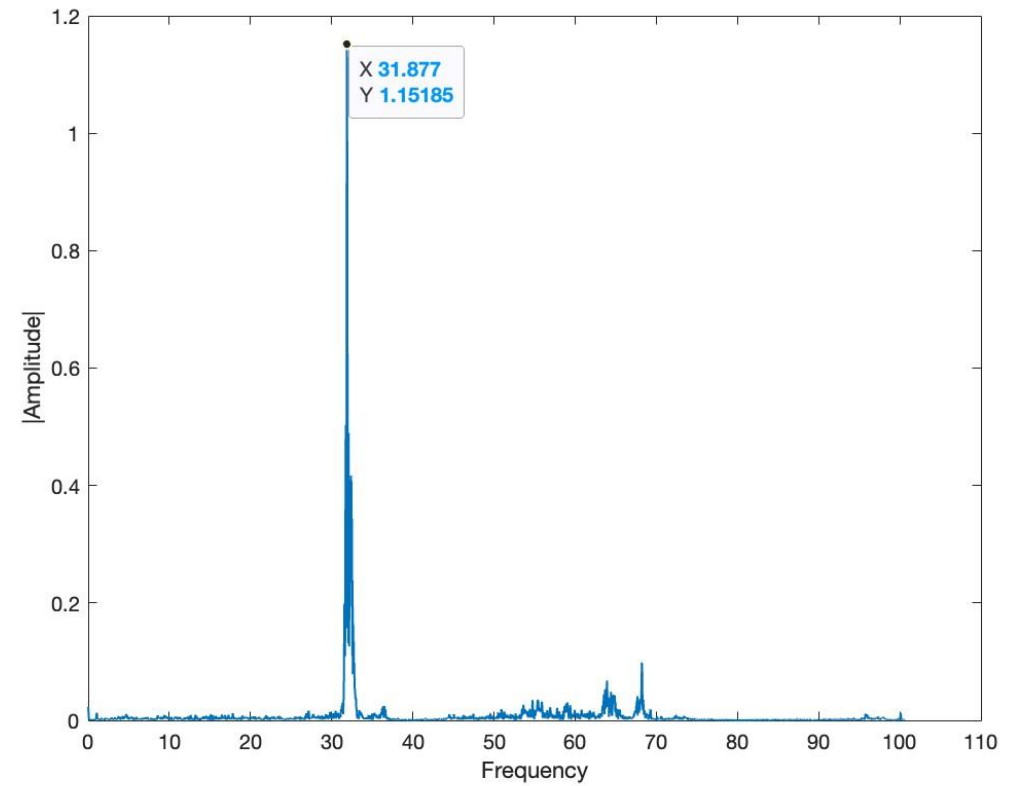


Fig : respective FFT plot

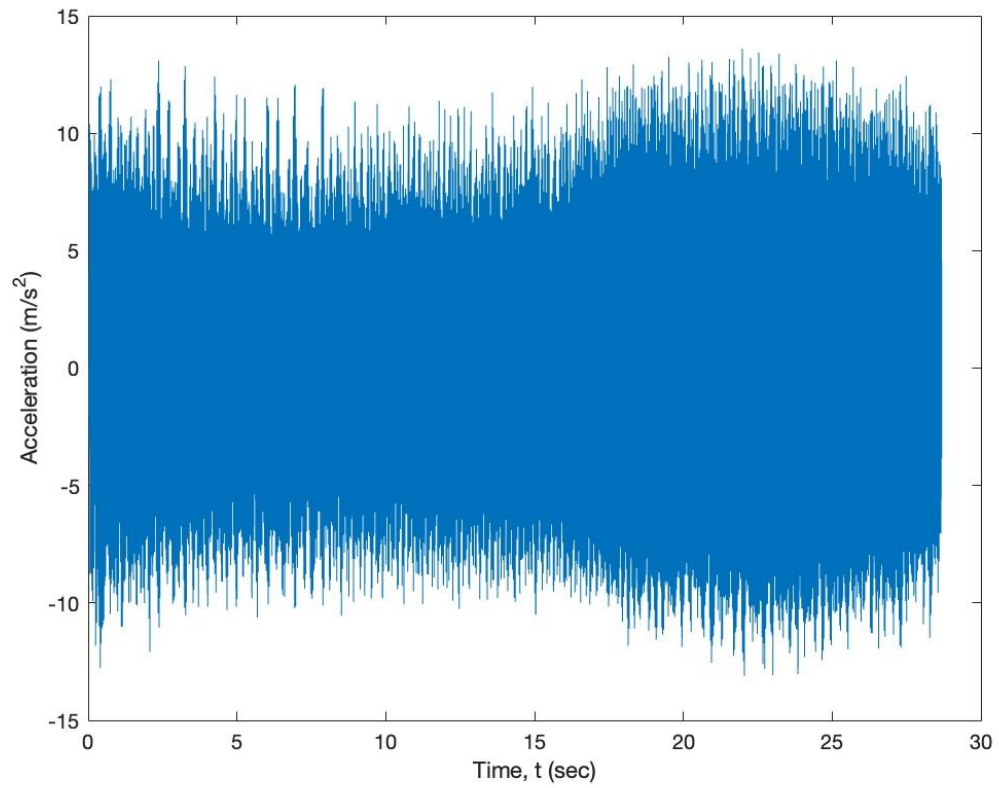


Fig : plot of Acceleration in y (m/s^2) Vs Time(sec)

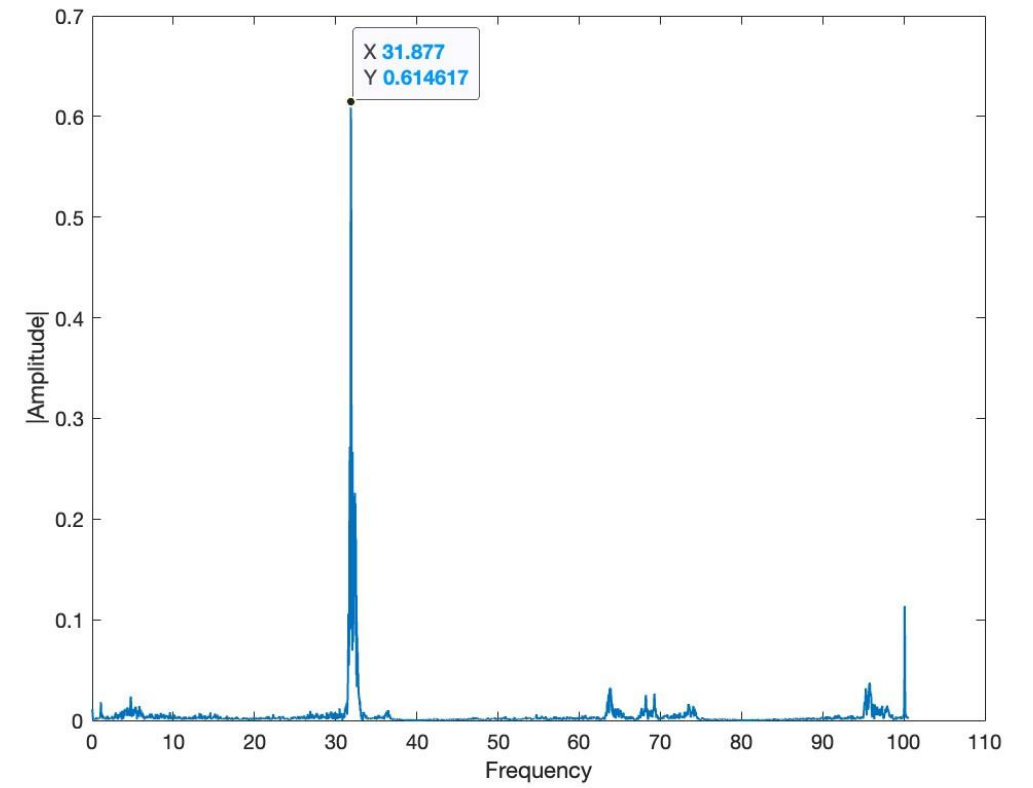


Fig : respective FFT plot

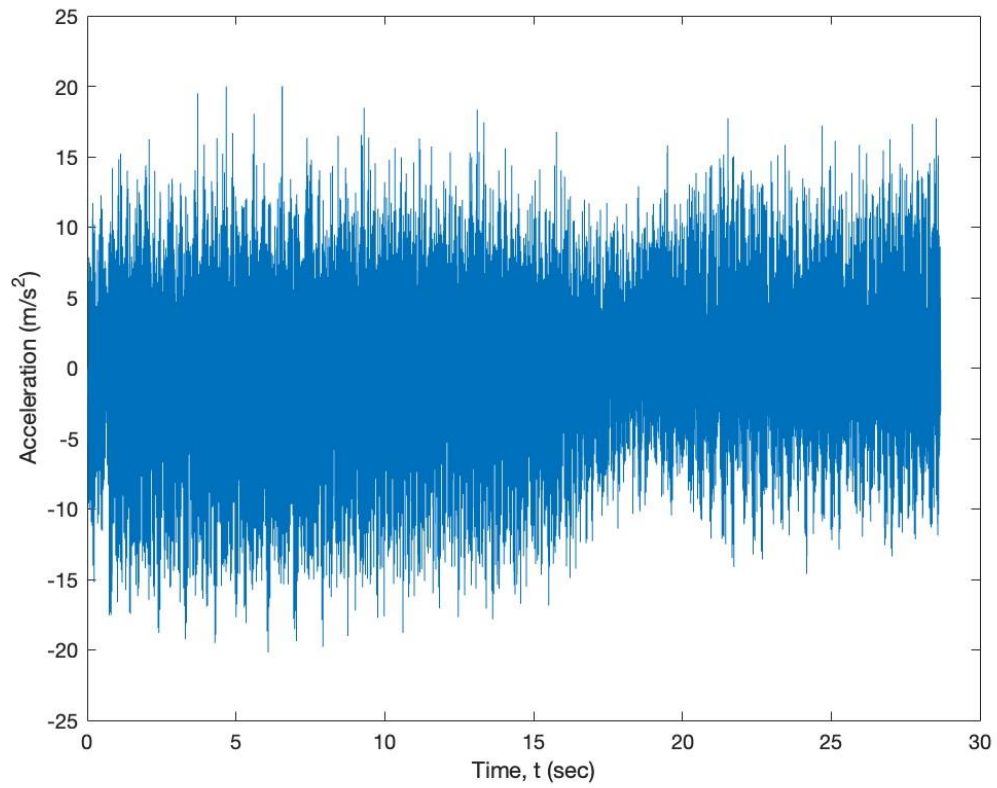


Fig : plot of Acceleration in z (m/s²) Vs Time(sec)

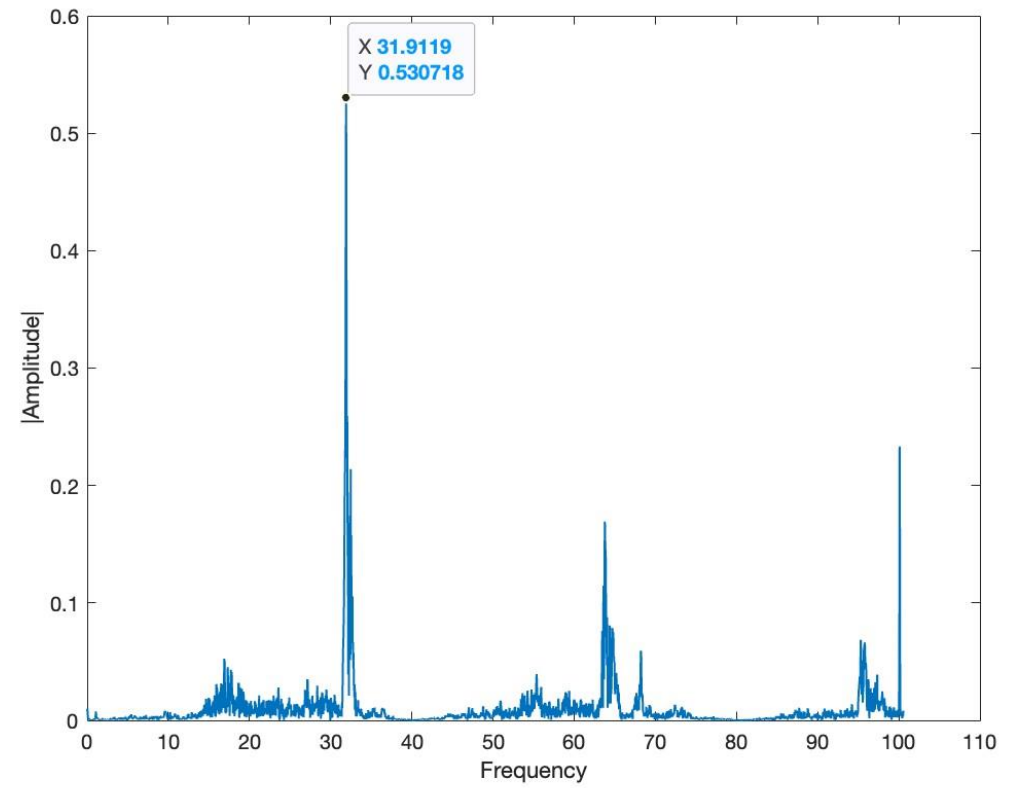


Fig : respective FFT plot

Balancing

- Removed mass (i.e opposite side blades)
- Due to this net force in x and y direction became zero (or forces canceled out each other)



Fig : blades

Comparison of FFT Plots at Speed 1

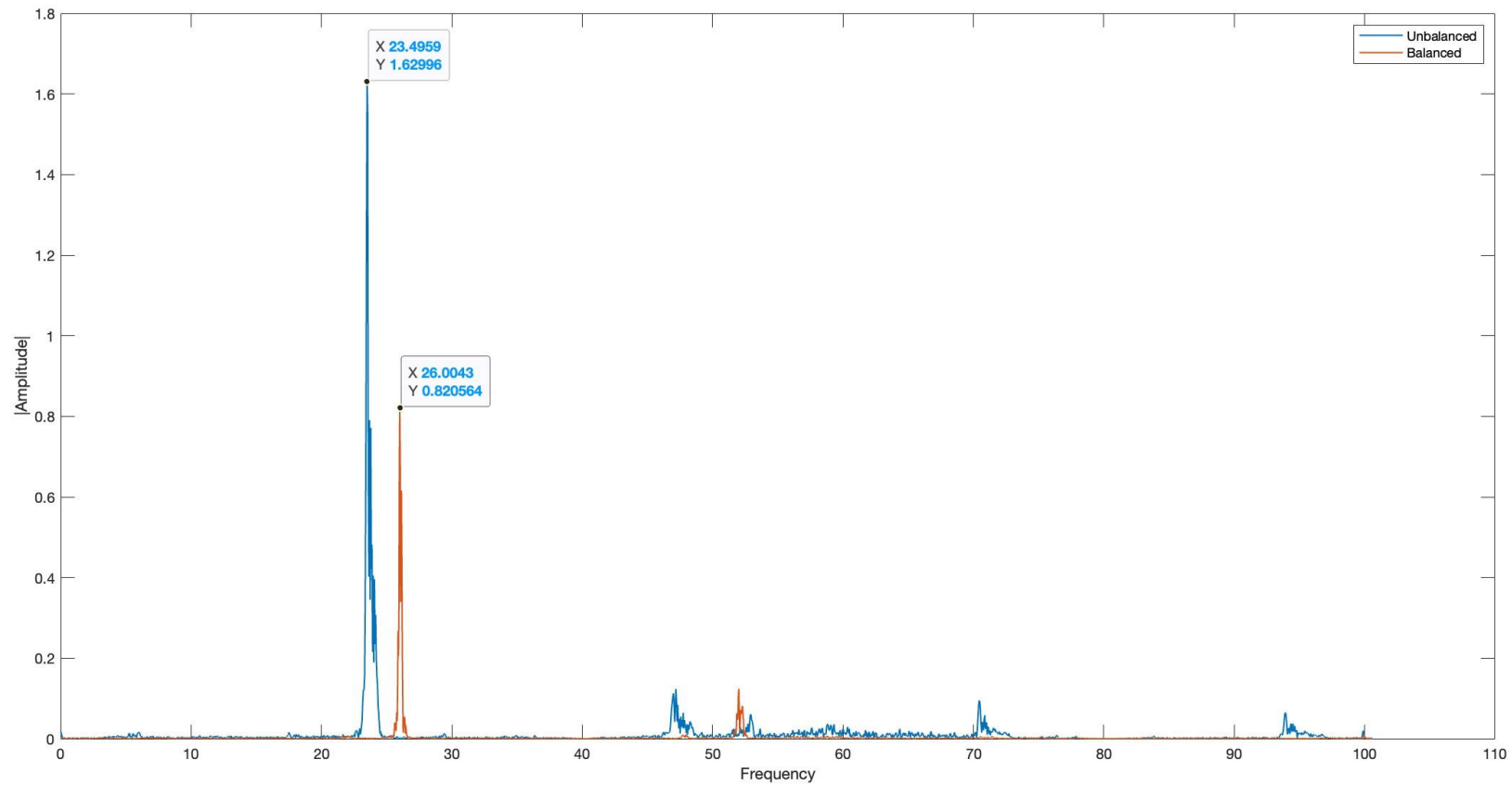


Fig : FFT plot in x-direction

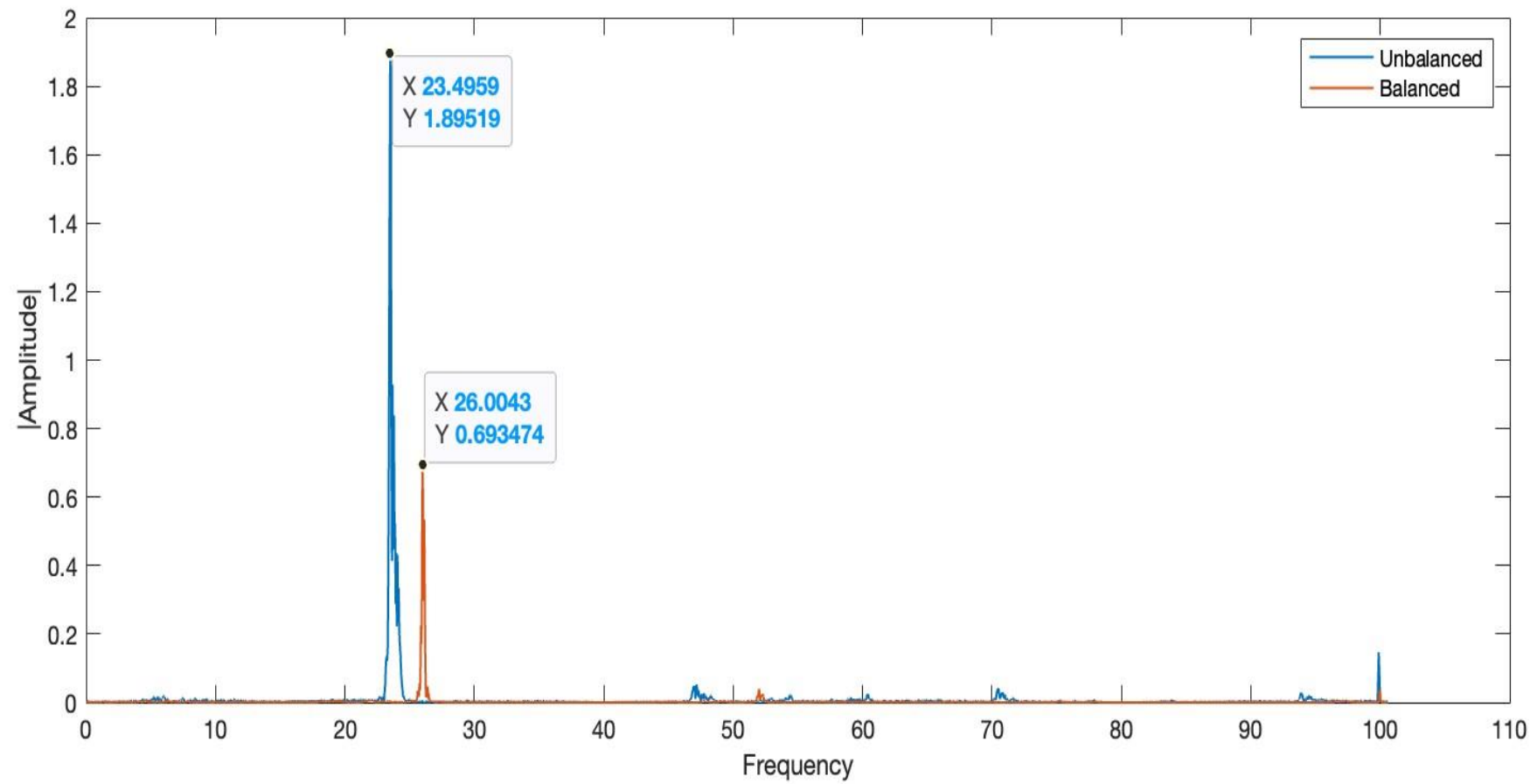


Fig : FFT plot in y-direction

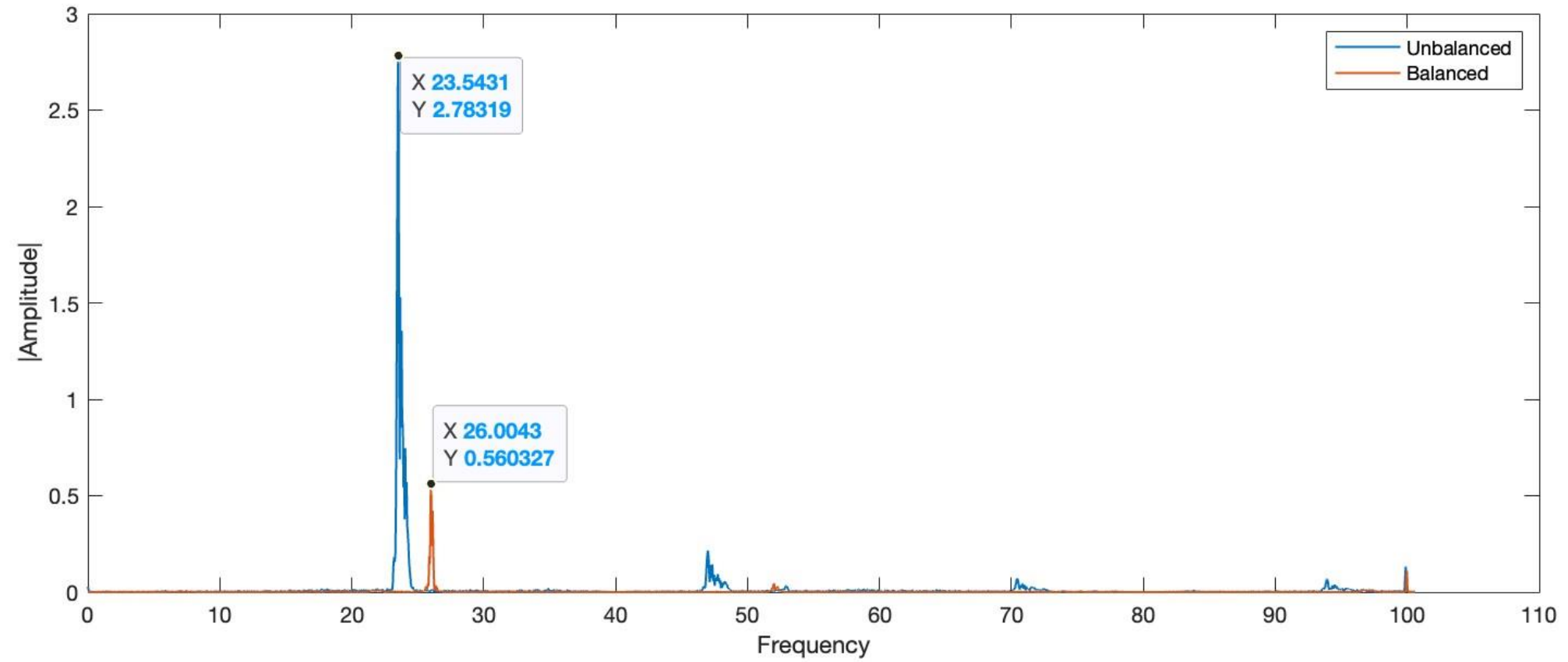


Fig : FFT plot in z-direction

Comparison of FFT Plots at Speed 2

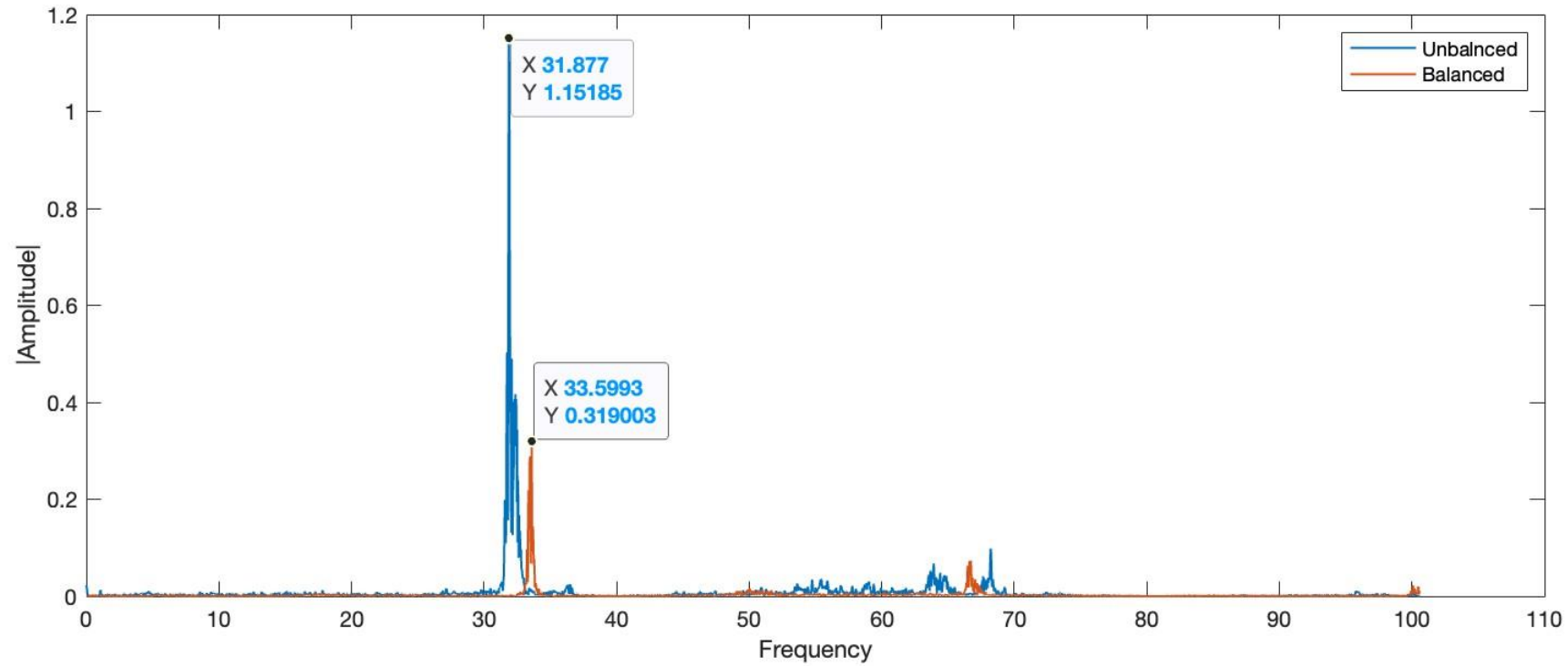


Fig : FFT plot in x-direction

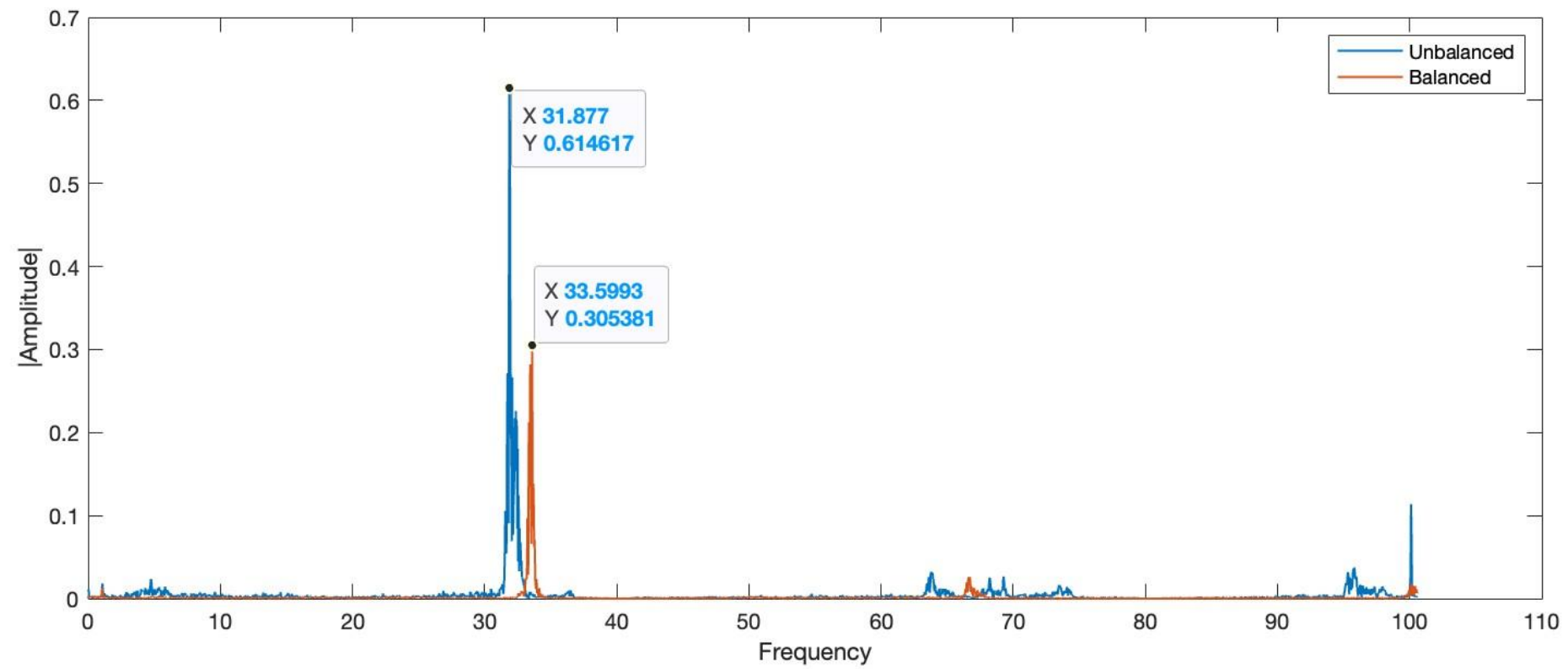


Fig : FFT plot in y-direction

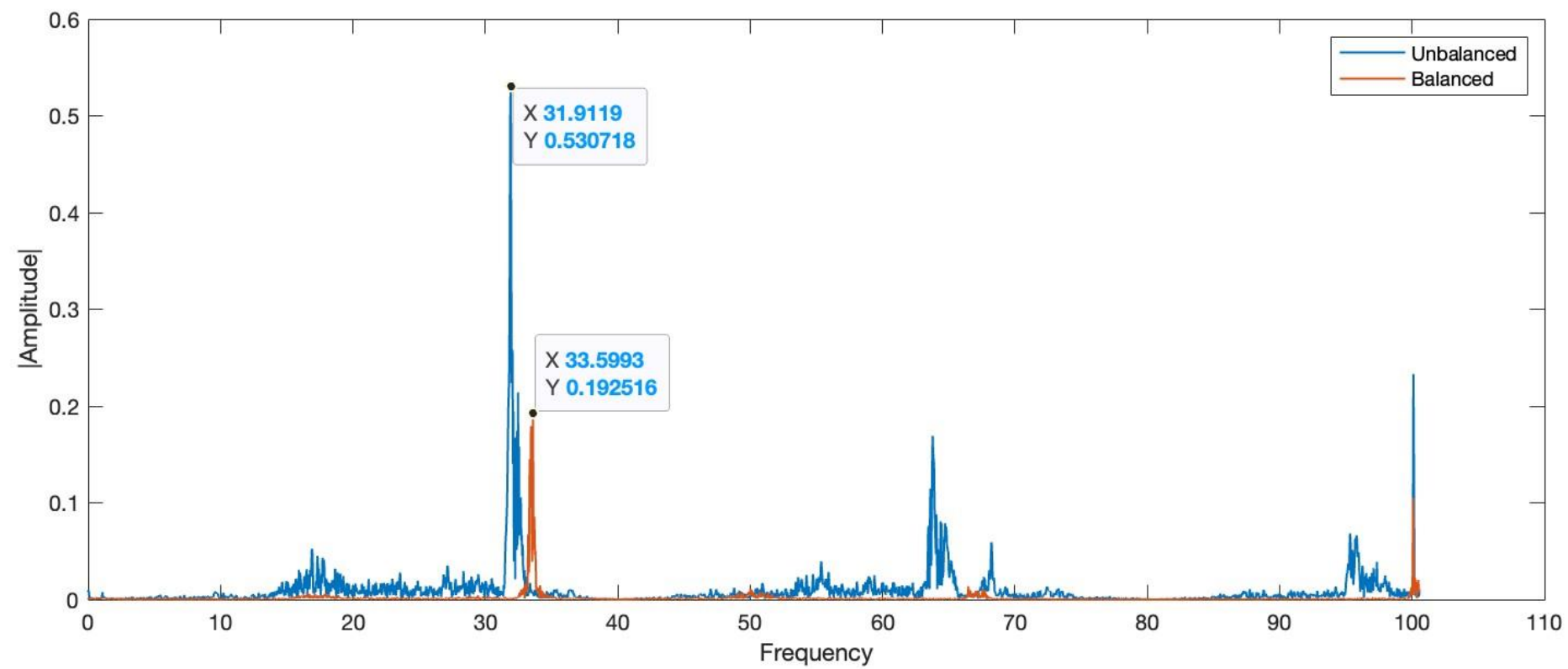


Fig : FFT plot in z-direction

Balanced Fan



Insights and conclusion

After balancing we got following Results :

- Stopped its circular motion
- Reduction in amplitude of vibration
- And increase life of cooling fan
- Noise reduction

THANK YOU!